CNN-Based Real-Time Driver Drowsiness Detection using Eye State Analysis

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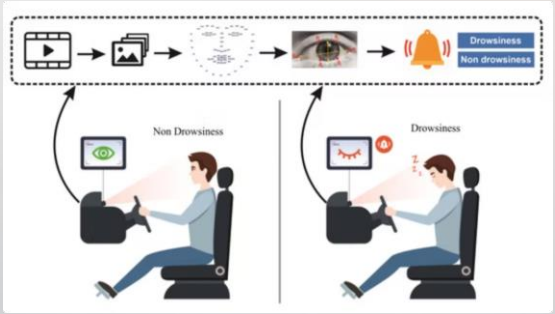
The Critical Need for Driver Alertness Monitoring

- Driver drowsiness causes **~20% of road accidents worldwide**
- Leads to **slow reaction time, poor judgment, and micro-sleeps (3–5 sec)**
- High risk during **long drives, night travel, and monotonous roads**
- Existing systems **lack real-time detection** or give **too many false alarms**
- Need for a **smart, reliable, real-time monitoring system**

Proposed Solution

- Detect driver drowsiness using **eye state (open/closed)**
- Focus on **eye region (ROI)** for accurate analysis
- Use:
  - **Convolutional Neural Networks (CNNs)** for classification
  - **Real-time video stream** for continuous monitoring
- System Workflow:
  - Capture video → Extract eye region → Classify state
- Alert Mechanism:
- If probability of drowsiness > 95
- If eyes remain closed for **> 300 ms**

System triggers **warning alarm**



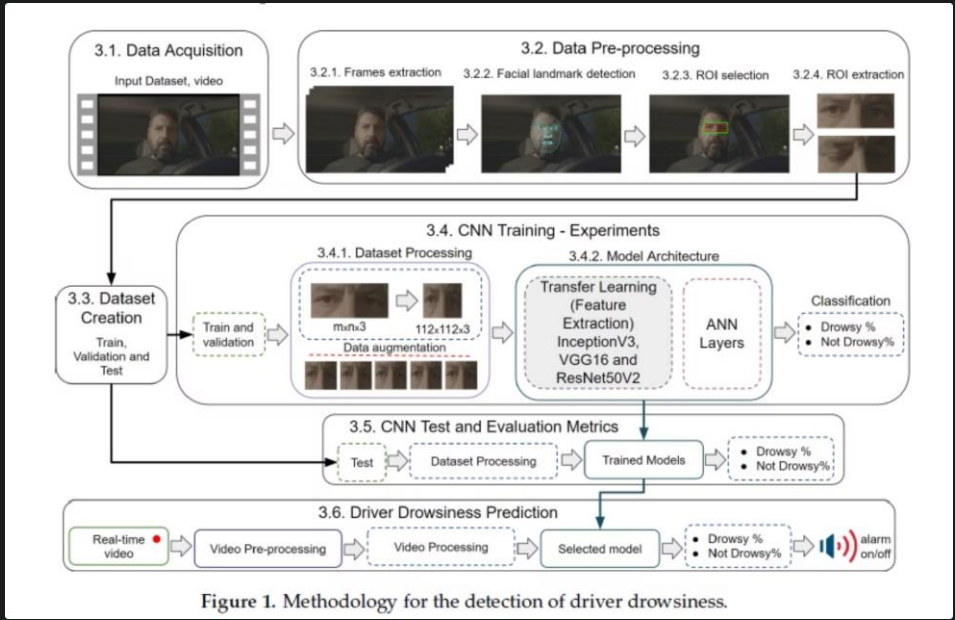
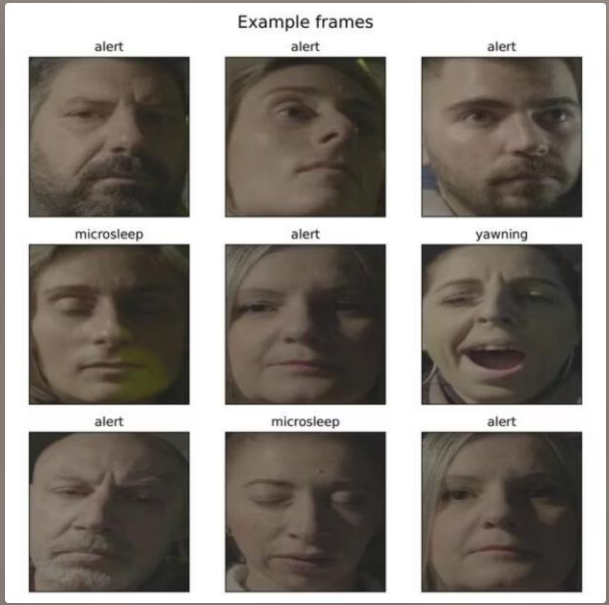


Figure 1. Methodology for the detection of driver drowsiness.

System Pipeline

The system works in 6 major stages:

1. **Data Collection:** Uses NITYMED dataset (real driving videos at night)
2. **Frame Extraction:** Videos → frames (25 FPS)
3. **Face & Eye Detection**: Uses MediaPipe Face Mesh Detects 468 facial landmarks
4. **ROI (Region of Interest) Extraction:** Focuses only on eye region Custom algorithm improves accuracy even when head moves
5. **CNN Training Models trained to classify:**
   - 😐 Not drowsy
   - 😴 Drowsy
6. **Real-Time Detection:**
   - If: Eye closed probability > 95% AND time > 300 ms → 🚨 Alert triggered



Dataset

- Dataset used: **NITYMED: https://datasets.esdalab.ece.uop.gr/download-files/**
- Consist of 130 Night-time driving videos
- Captured in **real driving conditions (night, fatigue)**
- Total Images: **30000**
- Classes:
  - **DROWSY**
  - **NOT DROWSY**
- Data Split:
  - **Training:** 80%
  - **Validation:** 20%

Model Architecture

- Used **Transfer Learning** with pre-trained CNN models:
  - InceptionV3
  - VGG16
  - ResNet50V2
- Custom Layers Added:
  - **Dense Layer:** 1000 neurons (ReLU)
  - **Dropout:** 30% (prevent overfitting)
  - **Output Layer:** Softmax (2 classes)
- Task:
  - Binary Classification → **Drowsy / Not Drowsy**

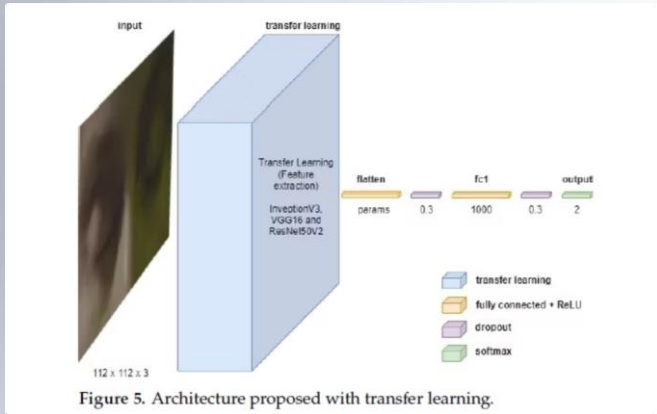
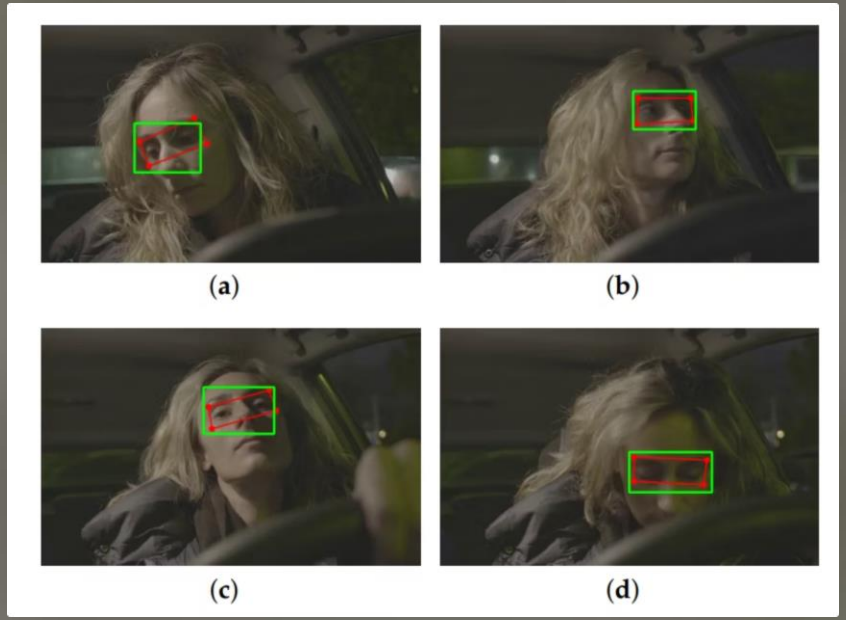


Figure 5. Architecture proposed with transfer learning.

Preprocessing

- **Frame Extraction**
  Convert real-time video into individual frames for analysis
- **Face & Landmark Detection**
  Use MediaPipe to detect **468 facial landmarks**
- **Eye Region Extraction (ROI)**
  Isolate eye area from landmarks for focused drowsiness detection
- **Image Resizing**
  Resize eye images to **112 × 112 pixels** for uniform input
- **Normalization**
  Scale pixel values to **0–1 range** for faster and stable training
- **Noise Reduction**
  Remove irrelevant background information to improve accuracy

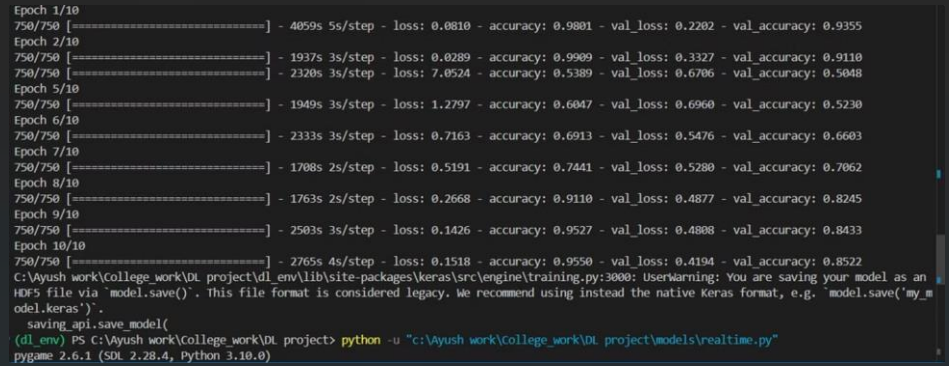
Helps in **reducing complexity, improving model performance, and ensuring consistent input**ons.



Training Details

- **Optimizer**
  Used **Adam optimizer** for efficient and adaptive learning
- **Epochs**
  Model trained for **30 epochs** to ensure proper convergence
- **Batch Size**
  Batch size of **32** for balanced training speed and performance
- **Loss Function**
  Used **categorical cross-entropy** for binary classification
- **Data Augmentation**
  Applied techniques like:
  - Rotation
  - Horizontal flipping
  - Zooming

    👉 Helps increase dataset diversity and prevent overfitting
- **Regularization**
  Used **Dropout (30%)** to reduce overfitting



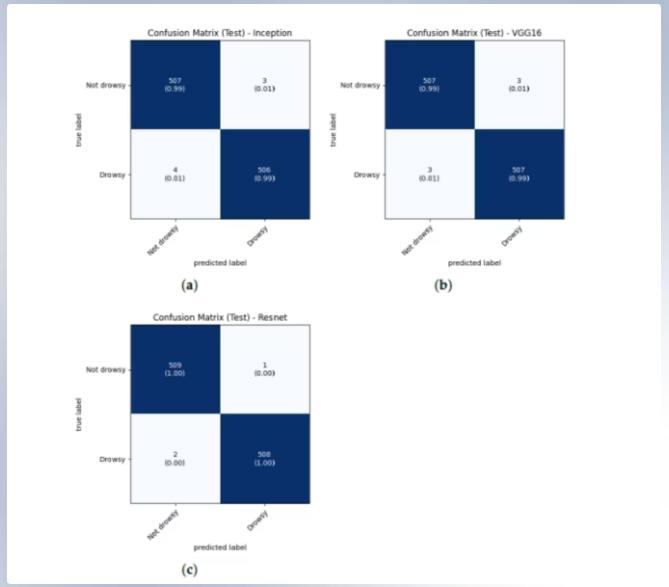
Evaluation Metrics

- **Accuracy**
  Overall correctness of the model
- **Precision**
  How many predicted drowsy cases are actually correct
- **Recall**
  Ability to correctly detect **actual drowsy cases**
- **F1-Score**
  Balance between precision and recall

Key Focus

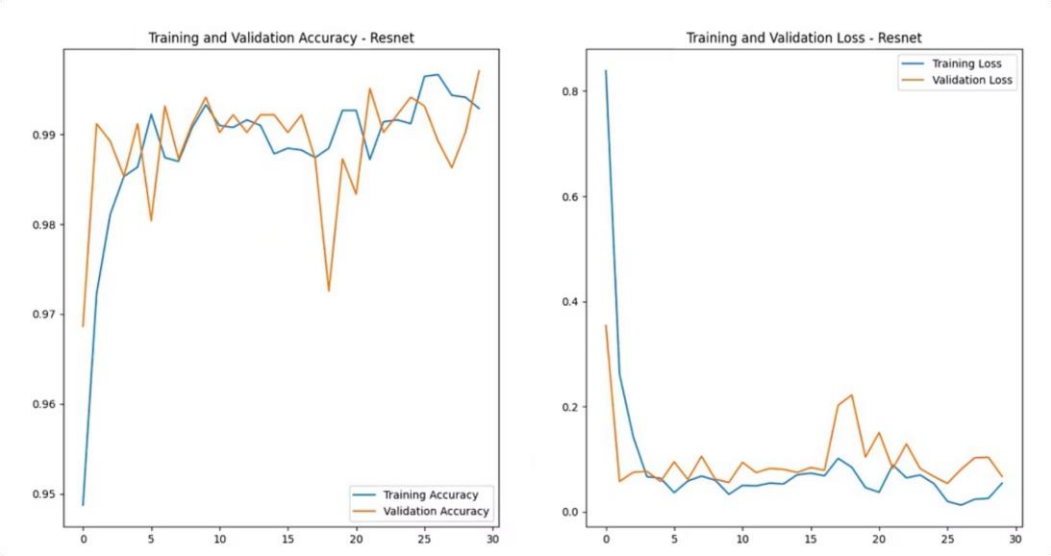
- **Minimize False Negatives**

(Missing a drowsy driver can lead to accidents 🚗)



Results

- Model performance evaluated on test dataset
- **Accuracy Achieved:**
  - InceptionV3 → **99.31%**
  - VGG16 → **99.41%**
  - ResNet50V2 → **99.71%**
- **Best Performing Model:**
  👉 **ResNet50V2** with highest accuracy
- **Observations:**
  - All models show **high accuracy (>99%)**
  - Transfer learning significantly improves performance
  - Model effectively distinguishes between **drowsy and alert states**
- **Inference:**
  System is **highly reliable for real-time drowsiness detection**



Visualization (Grad-CAM)

- Uses **Grad-CAM** for model interpretability
- Highlights **important regions** in the image used for prediction
- Model mainly focuses on:
  - **Eye region** 👁
  - Detecting open/closed state
- Generates **heatmaps**:
  - 🔴 High importance
  - 🔵 Low importance

Key Insight

- Helps understand **why the model predicts drowsy or not**
- Confirms that model is learning **relevant features (eyes)**

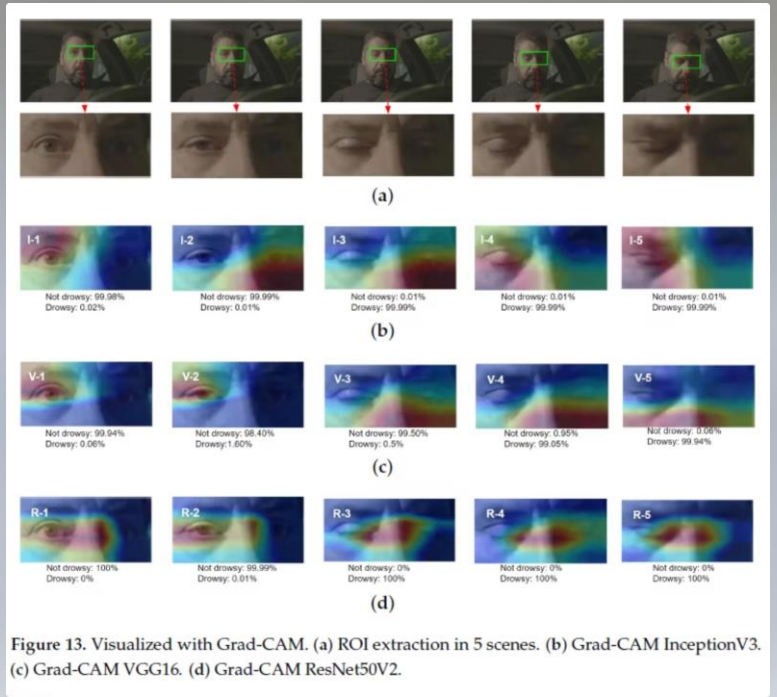
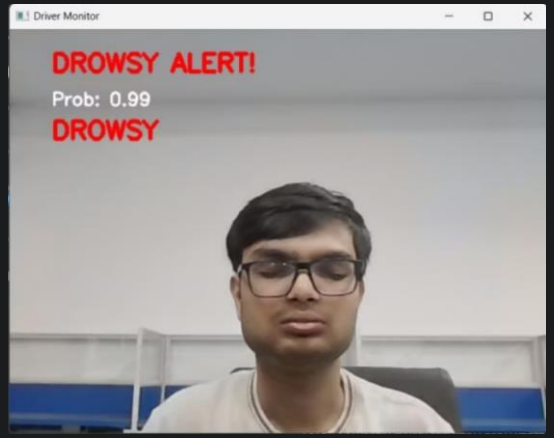


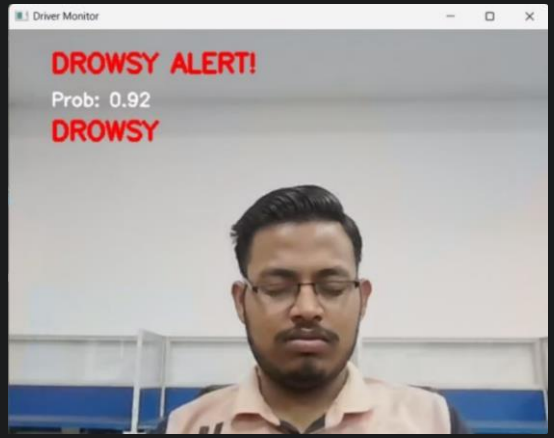
Figure 13. Visualized with Grad-CAM. (a) ROI extraction in 5 scenes. (b) Grad-CAM InceptionV3. (c) Grad-CAM VGG16. (d) Grad-CAM ResNet50V2.

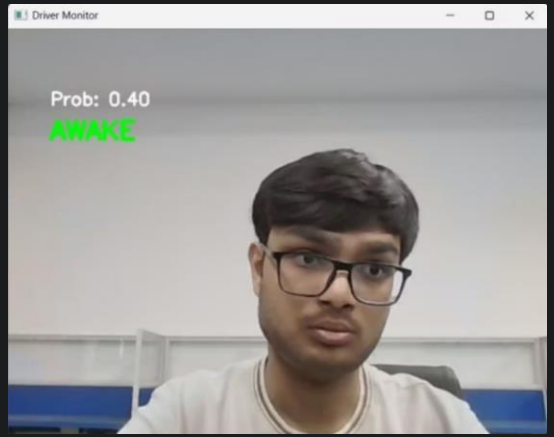
Real-Time Detection

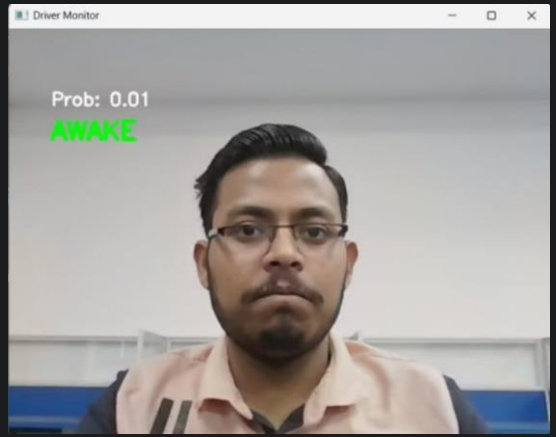
- Continuously captures and processes **live video stream** from the camera
- Detects face and extracts **eye region (ROI)** for each frame
- Uses trained **CNN model** to classify eye state as:
  - **Open (Alert)**
  - **Closed (Drowsy)**
- Tracks duration of eye closure across consecutive frames
- **Drowsiness Condition:**
  - If eyes remain closed for **> 300 ms**
  - System identifies driver as **drowsy**
- **Alert Mechanism:**
  - Triggers **audio alarm** 🚨
  - Provides instant warning to regain attention
- Displays real-time status: **Alert / Drowsy** on screen

👉 Ensures **continuous monitoring, quick detection, and immediate response for driver safety**


Driver Monitor
DROWSY ALERT!
Prob: 0.99
DROWSY


Driver Monitor
DROWSY ALERT!
Prob: 0.92
DROWSY


Driver Monitor
Prob: 0.40
AWAKE


Driver Monitor
Prob: 0.01
AWAKE

Advantages

- **High Accuracy (~99%)**
  Reliable and precise drowsiness detection
- **Non-Intrusive System**
  No need for wearable sensors or physical contact
- **Real-Time Monitoring**
  Continuous detection using live video feed
- **Robust Performance**
  Works even with **head movements & pose variations**

Limitations

- Performance depends on **lighting conditions** (low light may reduce accuracy)
- Requires **clear visibility of driver's face and eyes**
- May be affected by **occlusions** (glasses, hands, head movement)
- Accuracy can drop in **extreme real-world conditions**
- Requires **continuous camera access** for real-time monitoring

Future Work

- Use **Near-Infrared (NIR) Cameras**
  Improve performance in **low-light / night conditions**
- Add **Yawning Detection**
  Combine multiple fatigue indicators for better accuracy
- Deploy in **Embedded Systems**
  Integrate into **real vehicle environments**
- Enhance Model:
  - Improve robustness
  - Optimize for faster real-time performance

Conclusion

- Designed and implemented a **CNN-based driver drowsiness detection system** using eye state analysis
- Leveraged **transfer learning models (InceptionV3, VGG16, ResNet50V2)** to improve performance and reduce training time
- Achieved **very high accuracy (~99%)**, demonstrating strong model reliability
- **ResNet50V2** delivered the best results with highest accuracy and robustness
- System successfully classifies **drowsy and alert states** using real-time video input
- Detects prolonged eye closure (**> 300 ms**) to identify drowsiness effectively
- Provides **instant audio alerts**, helping the driver regain attention and avoid accidents

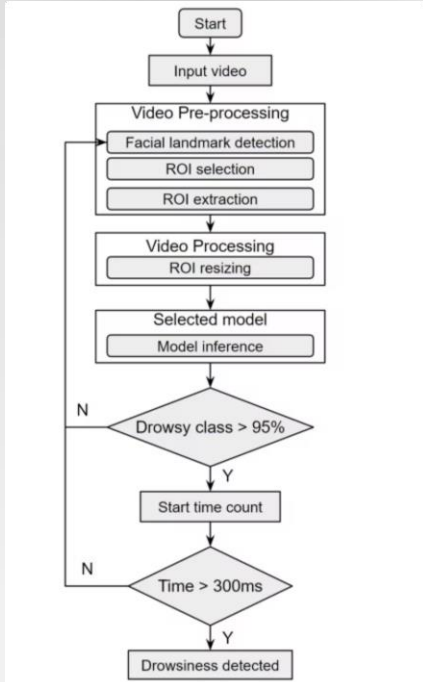


Figure 8. Drowsiness detection process flowchart.

THANK YOU

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